

Digit Test Process

Qualification criteria: Since the motherboards that we received were designed for specific processor and memory (RDRAM, DDR SDRAM, SDRAM) combinations we categorised the test process to include:

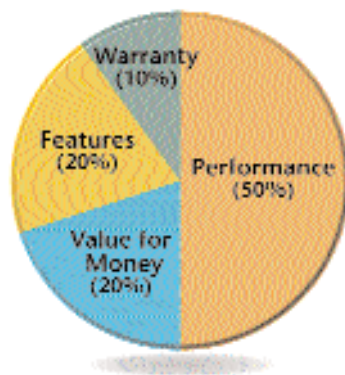
- Motherboards for Pentium4 (socket 478) supporting RDRAM, DDR SDRAM or SDRAM memory
- Motherboards for Pentium III / Celeron (Tualatin only) that used SDRAM memory
- Motherboards for AthlonXP / Duron (Palomino) supporting DDR SDRAM or SDRAM memory

Older motherboards that that were designed for the Pentium III (Coppermine) and Athlon (Thunderbird) processors were excluded from the test.

The test bench used the following common components:

nVidia GeForce3 64 MB DDR; 256 MB PC800 RDRAM or 256 MB PC2100 DDR SDRAM or 256 MB PC133 SDRAM (as applicable); Seagate Barracuda ATA IV 7200RPM Hard disk.

The best available hardware was used to test each motherboard type to ensure that the performance scores were not restricted by another component/subsystem.



Test methodology: The motherboards were evaluated on various parameters using a series of benchmarks. These included Performance (which was given a 50% weightage), the Features offered (20%), Value for money (20%) as well as the type of warranty offered (10%). The final rating of the motherboard, which is representative of its overall worth, was arrived at by putting together the scores in of each of the individual areas viz. of performance, features, value-for-money and the warranty offered.

Performance tests: Performance was evaluated using a series of benchmark programs that tested the performance of vari-

ous subsystems :

Processor subsystem: Using SiSoft Sandra 2002 Pro (Normal/Multimedia) and MadOnion's PCMark2002

Memory subsystem: Using PCMark2002 (Memory Index) and SiSoft Sandra 2002 Pro

Disk subsystem: Using SiSoft Sandra 2002 Pro (Drive Index) and HDtach 2.61

Graphics subsystem: Using Quake III Arena v 1.30 (Demo "four") and 3Dmark 2001SE

System-wide: Video Encoding using VirtualDub 1.4.7, Content Creation Winstone 2001 and Business Winstone 2001

Features: We awarded the motherboard points for the presence of features like ATA 133 and USB 2.0 support, the number of PCI and memory slots available, availability of on-board Ethernet or on-board 6-channel audio, support and other performance boosting features like BIOS supported overclocking.

Value for money: This score is derived taking into account the is a ratio of a product's Features and Performance to its price.

Warranty and support: We took into account the duration and other terms and conditions of the warranty offered (replacement/repair).

based ASUS A7N266-E performs brilliantly with DDR SDRAM.

The SDRAM solutions are best suited for Durons and lower speed Athlons. There is a world of difference between the performance of a KT133A or a KM133-based motherboard and the KT266A/KT333 motherboards. Our favourites are the KT333-based motherboards for high-speed Athlons and KT133A-based boards for Durons.

Memory matters

With the arrival of high-speed memory like DDR SDRAM and RDRAM over the last couple of years, choosing a motherboard has become more complicated.

Broadly speaking, if you are looking for a low cost solution then SDRAM is the way to go, even though all the tests show that this has a severe impact on performance, especially with the Pentium 4 and Athlon XP processors.

With the Pentium 4 processors, RDRAM is the fastest solution, but is quite expensive given the RAM and mother-

board prices. This is because RDRAM motherboards are manufactured using a six-layer fabrication process while the DDR boards use a less expensive four-layer process. For the best balance of price and performance, we recommend a DDR SDRAM option.

For the Athlon XP, the choice is easy. It works best with DDR SDRAM and the difference in performance when compared with SDRAM is considerable.

For Pentium III and Celeron processors, SDRAM is the only option available.

Performance numbers

The overall performance of a computer depends on performance of various specialised subsystems—the processor subsystem, the disk subsystem, the memory subsystem and the graphics subsystem. The motherboard is the foundation on which your entire system is built. A good motherboard takes advantage of every drop of performance that individual components (say, the graphics card) offer and ensures the smooth flow of data.

Processor subsystem

A motherboard is designed from chipset up to support a specific type or class of processor and its performance and feature set depends on the chipset. For instance, the Pentium 4 processor works differently than the Pentium III, thus the motherboard design is modified to take advantage of the higher clock speeds of the Pentium 4 and also the higher front side bus (FSB) frequency (400 MHz quad pumped). On the other hand, the Athlon XP processor requires an FSB speed of 266 MHz.

Pentium 4: As expected, the results varied according to the type of memory being used. However, there were a few surprises in store. The top performer in the Pentium 4 segment was not the Gigabyte GA8ITXE (using RDRAM) with a PCMark CPU index of 5307; it was outscored by the MSI 845 Ultra-C (using DDR memory) by a significant 24 points.

This is surprising, as you'd expect that Intel's heavy optimisations would have given the 850GB-based board an edge. →